

EE/CS 172/130
Digital Logic and Design

Topic 3 – Basic Logic Gates

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Logic Gates

- A **logic gate** is a physical device that performs a basic logical operation on one or more binary inputs to produce a single binary output.
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- The diagram illustrates a logic gate as a central rectangular box labeled "Logic Gate". To the left of the box, the word "Inputs" is positioned above two horizontal arrows pointing into the box. A vertical dotted line is placed between these two arrows. To the right of the box, a single horizontal arrow points away from the box towards the word "Outputs".
- The three basic logic gates are:
 - The AND gate
 - The OR gate
 - The NOT gate (inverter)
 - Some derived gates are:
 - The NAND gate
 - The NOR gate
 - The XOR gate
 - The XNOR gate
 - Logic gates are fundamental building blocks of a logic circuit or a digital systems.
 - Their name comes from their ability to perform logical decision-making: they produce one output level for certain input combinations and a different output for others.

Logic Functions

- A logic function is the mathematical relationship between input variables and the output, defined by Boolean algebra.
- Every logic gate implements a corresponding logic function.
- There are three different but equivalent ways to express and analyze the same logic function.
 1. Truth Table
 - A tabular representation of a logic function that lists all possible combinations of input variables along with the corresponding output.
 - It provides a complete description of the function's behavior.
 2. Logic Circuit
 - A graphical representation of a logic function using interconnected logic gates.
 - The circuit shows how inputs are processed through gates to produce the desired output.
 3. Logic Expression
 - An algebraic representation of a logic function using Boolean algebra symbols (such as AND, OR, NOT).
 - It provides a compact mathematical way to describe the same function.

The AND Gate

- Output is 1 iff all the inputs are 1.
- Alternatively, output is 0 if any of the inputs is 0.
- For a 2-input AND gate, let A and B be the two inputs and Z be the output, then:



- Logic expression: $Z = A.B$ or $Z = AB$

- Truth table:

A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

- Repeat for 3 inputs.

The OR Gate

- Output is 0 iff all the inputs are 0.
- Alternatively, output is 1 if any of the inputs is 1.
- For a 2-input OR gate, let A and B be the two inputs and Z be the output, then:



- Logic expression: $Z = A + B$

- Truth table:

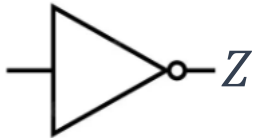
A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

- Repeat for 3 inputs.

The NOT Gate (Inverter)

- Has only one input and one output.
- Inverts / complements the input.
- Called NOT gate because the output is NOT the same as input.

- Let A be the input and Z be the output, then:

- Logic symbol: A  Z

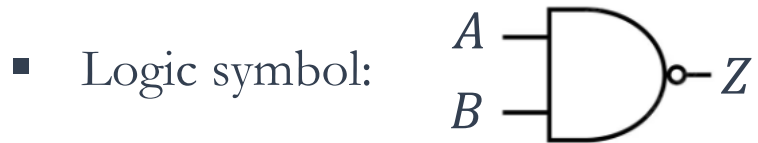
- Logic expression: $Z = \bar{A}$ or $Z = A'$

- Truth table:

A	Z
0	1
1	0

The NAND Gate

- Output is 1 if any of the inputs is 0.
- Equivalent to AND gate followed by an inverter.
- For a 2-input NAND gate, let A and B be the two inputs and Z be the output, then:



- Logic expression: $Z = \overline{A \cdot B}$ or $Z = \overline{AB}$

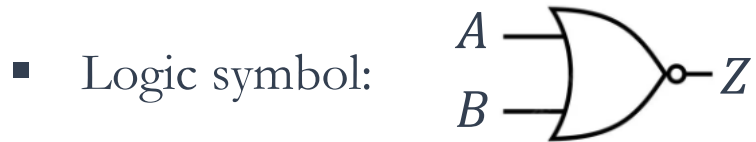
- Truth table:

A	B	Z
0	0	1
0	1	1
1	0	1
1	1	0

- Repeat for 3 inputs.

The NOR Gate

- Output is 0 if any of the inputs is 1.
- Equivalent to OR gate followed by an inverter.
- For a 2-input NOR gate, let A and B be the two inputs and Z be the output, then:



- Logic expression: $Z = \overline{A + B}$

- Truth table:

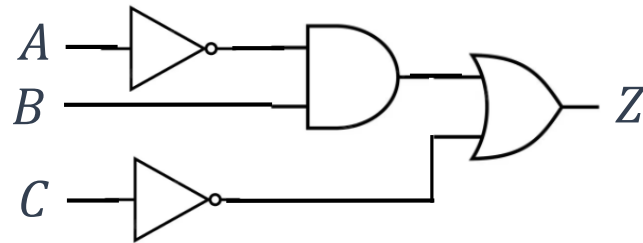
A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

- Repeat for 3 inputs.

Working with Logic Circuits

Finding Logic Expression and Truth Table for a given Logic Circuit

Example



$$Z = \bar{A}B + \bar{C}$$

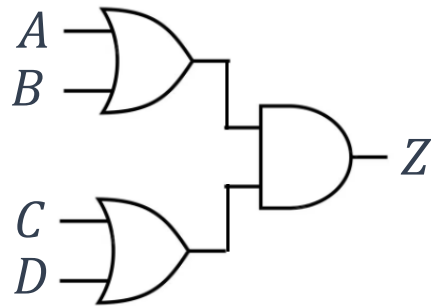
A	B	C	$\bar{A}$	$\bar{A}B$	$\bar{C}$	Z
0	0	0	1	0	1	1
0	0	1	1	0	0	0
0	1	0	1	1	1	1
0	1	1	1	1	0	1
1	0	0	0	0	1	1
1	0	1	0	0	0	0
1	1	0	0	0	1	1
1	1	1	0	0	0	0

Working with Logic Circuits

Finding Logic Circuit and Truth Table for a given Logic Expression

Example

$$Z = (A + B)(C + D)$$



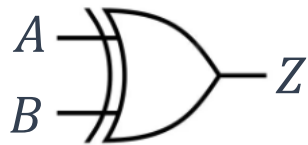
A	B	C	D	$A + B$	$C + D$	Z
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	1	1	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1

A	B	C	D	$A + B$	$C + D$	Z
1	0	0	0	1	0	0
1	0	0	1	1	1	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	0	0
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

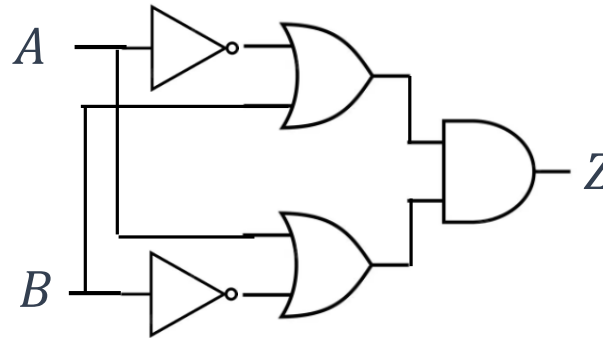
The Exclusive OR (XOR) Gate

For 2 inputs

- Output is 1 if both inputs are different.
- Let A and B be the two inputs and Z be the output, then:

Logic symbol:  $\equiv$

Logic expression: $Z = A \oplus B$
 $= \bar{A}B + A\bar{B}$



- Truth table:

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

The Exclusive OR (XOR) Gate

In general

- XOR gate is an odd function; output is 1 if the input variables have odd number of 1s.
- **Example:** For 3 inputs

<i>A</i>	<i>B</i>	<i>C</i>	<i>Z</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

- Repeat for 4 inputs.

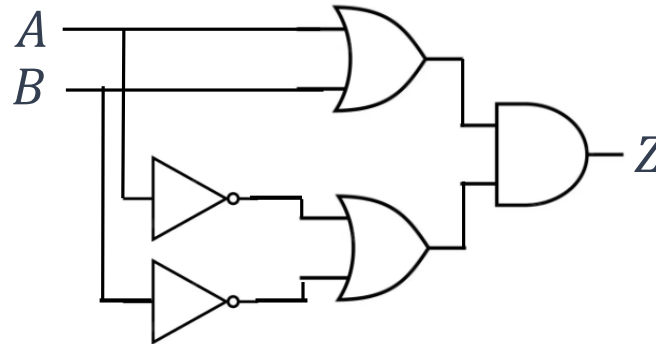
The Exclusive NOR (XNOR) Gate

For 2 inputs

- Output is 1 if both inputs are same.
- Let A and B be the two inputs and Z be the output, then:



$\equiv$



- Logic expression: $Z = A \odot B$
 $= AB + \bar{A}\bar{B}$

- Truth table:

A	B	Z
0	0	1
0	1	0
1	0	0
1	1	1

The Exclusive NOR (XNOR) Gate

In general

- XNOR gate is an even function; output is 1 if the input variables have even number of 0s.
- **Example:** For 3 inputs

<i>A</i>	<i>B</i>	<i>C</i>	<i>Z</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

- Repeat for 4 inputs.
 - For odd number of input variables, the outputs of XOR and XNOR gates are same.
 - For even number of input variables, the outputs of XOR and XNOR gates are complement of each other.

Problems

1. Each of the two temperature sensors produces a logical 1 when the temperature at its locations, T_1 and T_2 is above 20°C . A heater is to be turned off by a logical 1 received at its control unit when the temperature at either or both locations is above 20°C . Draw the required logic.
2. Each of the two temperature sensors produces a logical 0 when the temperature at its locations, T_1 and T_2 is above 20°C . A heater is to be turned off by a logical 1 received at its control unit when the temperature at either or both locations is above 20°C . Draw the required logic.
3. Each of the two temperature sensors produces a logical 1 when the temperature at its locations, T_1 and T_2 is above 20°C . A heater is to be turned off by a logical 0 received at its control unit when the temperature at either or both locations is above 20°C . Draw the required logic.
4. Each of the two temperature sensors produces a logical 0 when the temperature at its locations, T_1 and T_2 is above 20°C . A heater is to be turned off by a logical 0 received at its control unit when the temperature at either or both locations is above 20°C . Draw the required logic.
5. Each of the two temperature sensors produces a logical 1 when the temperature at its locations, T_1 and T_2 is above 20°C . A heater is to be turned off by a logical 1 received at its control unit only when the temperature at both locations is above 20°C . Draw the required logic.